# THE SET OF ELEMENTS FOR WHICH THE OBSTRUCTION TO DERIVED COMPLETENESS VANISHES IS A RADICAL IDEAL

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

*Editorial note.* The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

*Editorial note.* The author writes 'or because A_f \cong A_{f^n}' as the closing alternative, reproduced verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

*Allowed standard prerequisites (editorial).*

- The obstruction T(K, f) and its characterisation by the vanishing of Ext^n_A(A_f, K). invoked at proof lines 25942-25943. Converts the membership condition into an Ext vanishing condition that the ideal axioms can be checked against.
- The standard covering short exact sequence for two elements generating the unit ideal in a localization. used at proof lines 25946-25952. Is the sequence whose long exact Ext sequence gives additivity.
- Comparison of the localizations A_f and A_{f^n}, and T(K, f) = T(K, f^n). used at proof lines 25958-25959. Gives the radical property.

*Source reference.* Original lemma 00EJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 536–536. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 25850: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25850–25850. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

**Lemma 0.2.** *Let $A$ be a ring. Let $K \in D(A)$. The set $I$ of $f \in A$ such that $T(K, f) = 0$ is a radical ideal of $A$.*

*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* We will use the results of Lemma 25850 without further mention. If $f \in I$, and $g \in A$, then $A_{gf}$ is an $A_f$-module hence $\mathrm{Ext}^n_A(A_{gf}, K) = 0$ for all $n$, hence $gf \in I$. Suppose $f, g \in I$. Then there is a short exact sequence

$$0 \to A_{f+g} \to A_{f(f+g)} \oplus A_{g(f+g)} \to A_{gf(f+g)} \to 0$$

because $f, g$ generate the unit ideal in $A_{f+g}$. This follows from Algebra, Lemma 00EJ and the easy fact that the last arrow is surjective. From the long exact sequence of Ext and the vanishing of $\mathrm{Ext}^n_A(A_{f(f+g)}, K)$, $\mathrm{Ext}^n_A(A_{g(f+g)}, K)$, and $\mathrm{Ext}^n_A(A_{gf(f+g)}, K)$ for all $n$ we deduce the vanishing of $\mathrm{Ext}^n_A(A_{f+g}, K)$ for all $n$. Finally, if $f^n \in I$ for some $n > 0$, then $f \in I$ because $T(K, f) = T(K, f^n)$ or because $A_f \cong A_{f^n}$. $\square$